A Bayesian Standard Model of Early Word Learning

Joint inference across vocabulary, input, and processing data

Michael C. Frank

Stanford University

April 2026

Goal: model how language input drives vocabulary learning

A unified statistical model of early word learning, organized around four research questions:

RQ1. How well does word frequency reconstruct word difficulty?
i.e., is $\psi_j \approx -\log p_j$ *enough?*

RQ2. When does word learning start?
is start time s *identifiable from data?*

RQ3. Does vocabulary growth accelerate over development?
is δ *greater than 0?*

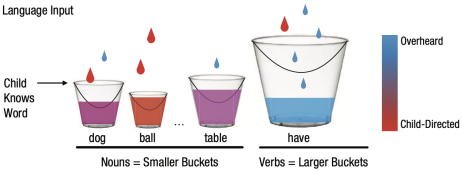
RQ4. How much of between-child vocabulary variance is driven by input rate vs. learning efficiency?
what is $\pi_\alpha = \sigma_\alpha^2/(\sigma_\alpha^2 + \sigma_r^2)$*?*

Follow-up to Kachergis, Marchman, & Frank (2021): "Toward a standard model of early language learning."

Schematic model: bucket-fill metaphor + item-response theory

Two ideas from Kachergis et al. (2021):

1. *Accumulator.* Each word is a bucket; child hears tokens at rate $r_i \cdot p_j$; efficiency α_i determines fraction retained.
2. *Item-Response Theory.* Per-word difficulty ψ_j controls how many "filled" tokens it takes for the child to know the word.



Model: linear predictor (anatomy)

$P(\text{produces}_{ijt} = 1) = \mathrm{logit}^{-1}(\eta_{ijt})$ where:

$$\eta_{ijt} = \underbrace{\lambda_j}_{\substack{\textit{item}\\ \textit{discrimination}\\ \textit{(2PL)}}} \left[\underbrace{\xi_i}_{\substack{\textit{child intercept}\\ = \log r_i + \log \alpha_i}} + \underbrace{\log p_j}_{\substack{\textit{word}\\ \textit{frequency}}} + \underbrace{\log H}_{\substack{\textit{waking}\\ \textit{hours/mo}}} + \underbrace{(1+\delta+\zeta_i)}_{\substack{\textit{1 + pop. accel.}\\ \textit{δ + per-child}\\ \textit{slope dev. ζ_i}}} \cdot \underbrace{\log \frac{t-s}{a_0}}_{\substack{\textit{log relative age}\\ \textit{(start time s)}}} - \underbrace{\psi_j}_{\substack{\textit{word}\\ \textit{difficulty}}} \right]$$

Factorization: ability vs. difficulty

The same equation regrouped as a 2PL IRT factorization:

$$\eta_{ijt} \;=\; \lambda_j \cdot \big(\, \theta_{it} \;-\; \beta_j \,\big)$$

Ability (child + age):

$$\theta_{it} = \xi_i + (1{+}\delta{+}\zeta_i)\,\log\tfrac{t-s}{a_0}$$

pop. accel. δ + per-child ζ_i

$$\xi_i \;+\; (1{+}\delta{+}\zeta_i) \;\cdot\; \log\tfrac{t-s}{a_0}$$

child intercept $\log r_i + \log \alpha_i$

start time s

Difficulty (item):

$$\beta_j = \psi_j - \log p_j - \log H$$

per-word log threshold tokens (class-hierarchical)

$$\psi_j \;-\; \log p_j \;-\; \log H$$

corpus frequency (unit coefficient)

hours/month (constant)

Why split it. $\log p_j$ and $\log H$ together convert "probability" into "expected tokens per month at unit

Model variants

Per-child latents. $(\log \alpha_i, \zeta_i) \sim \mathrm{MVN}(0, \Sigma)$ with LKJ prior on the correlation; $\log r_i \sim N(\mu_r, \sigma_r)$ independent. *Sum-to-zero centering on each random effect* identifies δ (population slope) separately from $\mathrm{mean}(\zeta_i)$.

Lean baseline + opt-in variants. Pin $s = 0$, $\sigma_\lambda = 0$, $\sigma_\zeta = 0$ by default; opt in explicitly:

variant	adds
`slopes`	ζ_i, σ_ζ (longitudinal default)
`2pl`	λ_j, σ_λ
`free_s`	free s
`fix_delta`	$\delta = 0$

σ_r **pinned externally** from Sperry / Hart-Risley / Weisleder-Fernald data; lets the model decompose $\sigma_\xi^2 = \sigma_r^2 + \sigma_\alpha^2$ and report π_α.

Main longitudinal sample: Wordbank English + Norwegian

Data. Wordbank longitudinal CDIs (WG $\cup$ WS forms). Children with ≥ 2 admins; both forms count toward the same child.

	N children	N admins	age range (mo)
English (American)	200	$\sim$808	8–32
Norwegian	200	$\sim$1 594	8–32

Stratified subsampling.

- **Children**: 4 age-bins on median admin age, equal n per bin; favor those with most admins / longest spans.
- **Items**: 4×4 stratification on (lexical class $\times$ $\log p_j$ quartile). 200 items kept.

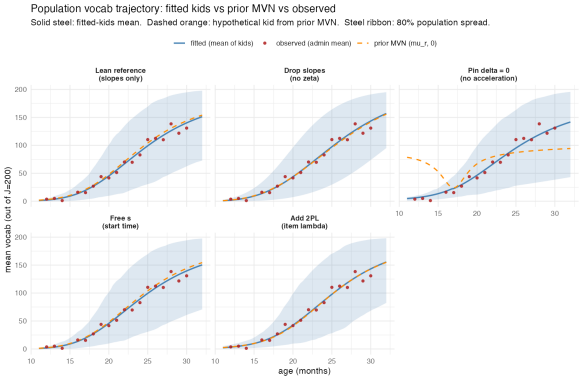
Stan. `log_irt_long.stan`; 4 chains $\times$ 1000 iter $\times$ 500 warmup; adapt_delta $= 0.95$; runs in 3–8 hours on Sherlock.

Results: lean baseline (English, slopes only)

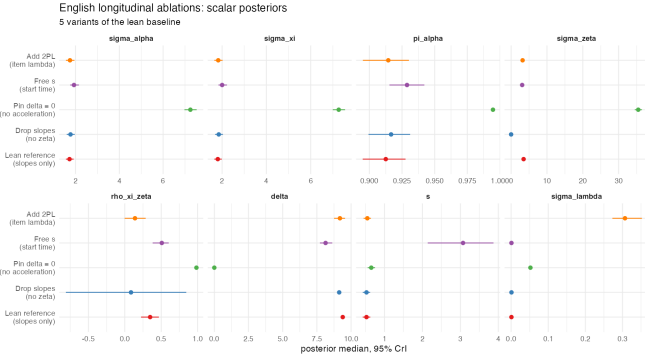
Headline scalars (post. med. [95% Crl]):

Population		
σ_α	1.83	[1.65, 2.04]
σ_ξ	1.90	[1.73, 2.11]
σ_ζ	3.48	[3.05, 3.96]
$\rho_{\xi\zeta}$	0.40	[0.27, 0.53]
δ	9.39	[8.58, 10.2]
s	0.52	(pinned)
$\boldsymbol{\pi_\alpha}$	**0.913**	**[0.901, 0.926]**

Efficiency variance dominates input variance: $\pi_\alpha \approx 0.91$.



Ablations: what each component buys you



Pin $\delta = 0$: population MVN no longer matches data; $\sigma_\zeta \to 35$ trying to compensate. δ **is load-bearing.**

Drop slopes: population fit preserved; slopes matter for per-child trajectories, not population mean.

Free s: s wanders, other params little affected.

Add 2PL ($\sigma_\lambda \approx 0.31$): item discrimination real but population-averaged-out.

Why we want input-uptake datasets

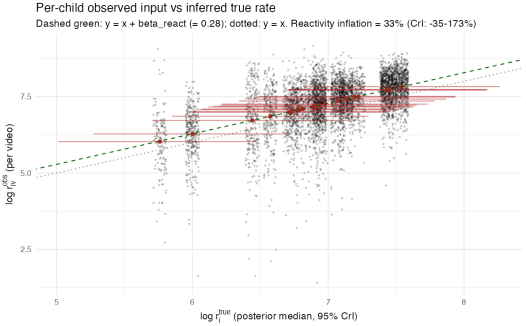
The longitudinal fit **infers** σ_r from external Sperry-style priors. To *measure* between-child input variance directly, we need datasets with per-child observed input rates.

	BabyView (Long et al. 2025)	**SEEDLingS** (Bergelson; Egan-Dailey 2025)
Recording channel	head-mounted body camera + transcript	passive LENA all-day audio
Observer effect	**yes** (β_{react} free)	**no** (β_{react} pinned at 0)
N kids (post-link)	20	44
CDI form	WG $\cup$ WS, longitudinal	WG, 6–18 mo
Per-recording rate	$\log r_v^{\text{obs}}$ from FEM/MAL tokens	$\log r_v^{\text{obs}}$ from LENA AWC

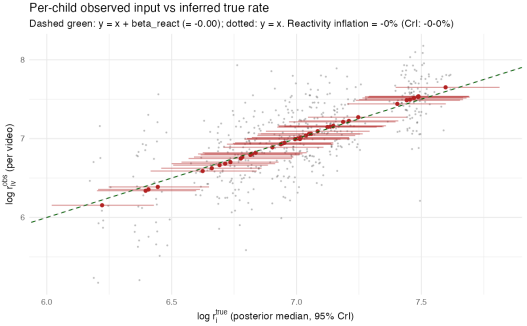
Modeling. Same Stan model (`log_irt_io.stan`); β_{react} on/off via dataset-level prior.

IO results: π_α replicates across samples

BabyView io_slopes (N=20)



SEEDLingS io_slopes (N=44)



dataset	N kids	π_α	δ	σ_α
English (long_slopes)	200	0.91 [0.90, 0.93]	9.39	1.83
BabyView (io_slopes)	20	0.86 [0.73, 0.94]	6.40	1.20
SEEDLingS (io_slopes)	44	**0.99 [0.99, 1.00]**	8.04	2.59

Three samples, two input-observation methods, one unobserved-input baseline: efficiency variance dominates.

Peekbank experiment: LWL as a window onto $\log \alpha$

Sample. 62 Adams & Marchman (2018) children with both LWL RT/accuracy from Peekbank (247 admins, 13–20 mo) and item-level CDIs at 22 + 25 mo from the Stanford TotLot 3 cohort.

Joint model.

$$\log \mathrm{rt}_{i,t} \sim \mathcal{N}\big(\mu_{rt} + \mu_{\mathrm{rtslope}}\,\ell_t \;-\; \boldsymbol{\gamma_{\mathrm{rt}}}\,\log \alpha_i \;+\; \ell_t\,\mathrm{rtslope}_i,\; \sigma_{\mathrm{lwl}}\big)$$

where $\ell_t = \log(t/a_0)$. $(\log \alpha, \zeta, \mathrm{rtslope}) \sim \mathrm{MVN}$ with LKJ prior on the correlation matrix.

Result.

- $\boldsymbol{\gamma_{\mathrm{rt}} = 0.073}$ $[0.032, 0.117]$ – bounded away from zero. Each unit of $\log \alpha$ corresponds to ~12% lower mean RT.
- $\mu_{\mathrm{rtslope}} = -0.73$: RT *decreases* with age, expected.
- $\sigma_{\mathrm{rtslope}} = 0.74$: real per-child variation in RT-improvement rate.

Comparison to prior accumulator models

Model	Per-child	Per-word	Acceleration	Bayesian	Real units
McMurray 2007	–	$g(\tau)$	via g'	–	–
Mitchell & McMurray 2009	–	$g(\tau)$ + leverage	via g'	–	–
Hidaka 2013	–	(N, δ, D)	via D	–	–
Mollica & Piantadosi 2017	–	(k_w, λ_w)	–	✓	partial
Kachergis et al. 2021	θ_i	d_j, class	–	–	–
Ours (M5)	$\xi_i, \zeta_i, \pi_\alpha$	$\psi_j, \lambda_j, \beta_c$	δ free	✓	✓

What's new in our nested family.

- McMurray 2007 / Mitchell & McMurray 2009: deterministic, no per-child variation. Their $g'(\tau) > 0$ requirement maps onto our **M2 vs. M1** test (does $\delta > 0$?).
- Hidaka 2013: closed-form Gamma / Weibull AoA distributions, population-averaged. Our **M1, M2, M3** occupy the same family (cumulative, rate-change, both) at the per-(child, word, age) Bernoulli level.
- Mollica & Piantadosi 2017: (k_w, λ_w) Gamma waiting time; large-k approximated by our logistic. Their joint (k, λ) ridge is our (s, δ) ridge.
- Kachergis et al. 2021: standardized logit IRT. Our model adds external σ_r pin (RQ3), structural δ (RQ1), real token units, and observation-channel architecture (uptake / processing).

Conclusions

- $\pi_\alpha \approx 0.86$–0.99 *across three samples*: between-child vocabulary variance is driven primarily by efficiency, not input rate.
- $\delta \approx 6$–9 *across all variants*: vocabulary growth genuinely accelerates with age; pinning $\delta = 0$ breaks population fit.
- LWL processing speed informs $\log \alpha$ in the expected direction; coupling $\gamma_{rt} > 0$ is bounded away from zero.

Code, data, fits, and full explainer: `github.com/langcog/standard-model-2`